Abstract

Standardized test scores are closely tied to student's socioeconomic circumstances, but it is not clear how much of the variation in school's average scores can be explained by them, type of school, and location. This project combined the EdGap data set (EdGap, 2017) with National Center for Education Statistics school information (National Center for Education Statistics, 2017) for 7,227 high schools in 20 states and fit multiple linear regression models predicting average ACT score using socioeconomic measures, state, school type, and charter status. Models were compared using ANOVA, RMSE, MAE, and R-squared. The final model, which used log median income and excluded school type and charter status, explained about 68% of the variation in school's average ACT score. Free and reduced-price lunch had the largest association (-1.82 points per standard deviation), followed by percent of adults with a college degree (0.40). There were noticeable differences in scores based on state, and school type and charter status added no detectable explanatory value. The results describe associations in observational data.

Introduction

ACT and SAT scores are some of the most widely used indicators of academic performance for high schoolers and play a large role in their college admissions. Research shows that these scores are closely tied to the circumstances in which students live. A 2025 paper from Opportunity Insights, a Harvard-based team of researchers and policy analysts, found that "Test scores differ sharply by parental income, with children from high-income families having much greater chances of scoring at the top of the distribution than those from lower-income families" (Chetty et al., 2025, p. 13). Similarly, Geiser found family income, parental education, and race/ethnicity together play a huge role: "40% of the variation in students' SAT/ACT scores is attributable to differences in socioeconomic circumstance" (Geiser, 2020, p. 12). These findings describe individual students. If the relationship holds across schools, then a school's average score may reflect its students' circumstances as much as its instruction.

Two data sets were combined for the 2016-2017 school year. The first is the EdGap data set, which takes socioeconomic data from the Census Bureau's American Community Survey and ACT scores from each state's department of education or some other public data release. It contains 7,986 schools identified by their NCESSCH school ID. For each school, it provides the average ACT score (or SAT equivalent), the percent of students receiving free or reduced-price lunch, and four measures of socioeconomic status including unemployment rate, percent of adults with a college degree, percent of children living in a married-couple family, and median household income. The second is school information data from the National Center for Education Statistics. This data set consists of 102,183 records of basic identifying information about schools, from which we keep the school ID, state, zip code, school type, school level, and charter status.

The two data sets were merged on school ID, keeping every record from the EdGap data set. Numerical variables were then checked against their bounds. An ACT score below one and negative lunch percentages are impossible, so they were set to missing. The data were then restricted to high schools. Three schools with missing average ACT scores were dropped, leaving 7,227 schools from 20 states. The remaining missing values were estimated with an iterative imputer, which models each variable using the others.

The scientific aim of this project is to quantify how much of the variation in school's average ACT scores can be explained by state, school type, charter status, and other socioeconomic factors, and to identify which of these factors are the most important predictors when the others are held constant. These questions are addressed with multiple linear regression models.

Theoretical Background

Multiple linear regression. The analysis models a school's average ACT score as a linear function of several predictors: $ACT_i = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + \ldots + \beta_k x_{ki} + \varepsilon_i$

Coefficients are estimated using ordinary least squares, which minimizes the sum of squared residuals. The coefficients are interpreted as the expected change in ACT score for a one-unit increase in that predictor, holding the others constant. Residual plots are used to check for non-linearity and non-constant variance, which violate the model's assumptions of a linear relationship and constant error variance.

Categorical predictors: State, school type, and charter status are categorical variables, so each is encoded as indicator variables, with one category left out as a reference. Each coefficient measures the difference from the reference category.

Transformations: Median income is right-skewed. Using log(median income) reduces the skew and makes the coefficient reflect the change in ACT score based on a percentage change in median income.

Standardizing: Numeric predictors are made to have a mean of 0 and a standard deviation of 1 by subtracting the mean and dividing by the standard deviation. This does not change the model's fit but puts coefficients in terms of standard deviations so the effect size can be compared across predictors.

Missing data: Iterative imputation is used to estimate each missing value by regressing that variable on the others. This is reasonable because there are few missing values and the predictors are correlated.

Model comparison: Models are compared using root mean squared error (RMSE), mean absolute error (MAE), and R-squared. ANOVA is used to compare full models with reduced and asks whether the additional predictors reduce the residual sum of squares by more than is expected by chance. A large p-value means the extra predictors do not significantly improve the fit, so the simpler model is preferred.

Methodology

All analysis was done in Python (pandas, scikit-learn, statsmodels, matplotlib, seaborn).

The data were prepared by merging and cleaning the EdGap and school information as described in the introduction. After restricting the data to high schools and checking for duplicate rows, schools with missing average ACT scores were dropped. The remaining missing values in the five numeric predictors (unemployment rate, percent college, percent married, median income, and percent lunch) were imputed with scikit-learn's iterative imputer with default settings. The final data set contained 7,227 schools with no missing values.

Multiple linear regression models using ordinary least squares were fit using statsmodels with numerical predictors and state, school type, and charter status as categorical variables. State uses Delaware as its reference category, school type uses Alternative School, and charter uses non-charter.

Four models were fit:

Model	Predictors
Full	Unemployment rate, percent college, percent married, median income, percent lunch, state, school type, charter
Reduced	Unemployment rate, percent college, percent lunch, state
Full log income	Unemployment rate, percent college, percent married, log median income, percent lunch, state, charter
Reduced log income	Unemployment rate, percent college, percent married, log median income, percent lunch, state

Variables were removed to form each reduced model when they were not significant, using alpha = 0.05, in the full model. School type was left out of the log models because it's only significant category in the full model had 1 observation. Each model's residuals were then plotted against its fitted values to check for non-linearity and non-constant variance.

Each reduced model was also compared against its full model using ANOVA. For a non-significant result, the simpler model was preferred. All models were then compared using RMSE, MAE, and R-squared. The final model was the one that performed significantly better in these tests.

The numeric predictors in the final model, Reduced log income, were standardized to have a mean of 0 and a standard deviation of 1 then fit to a new model.

Computational Results

Four regression models were fit to the 7,227 schools. Table 1 gives the nested-model comparison, Table 2 summarizes their fit, Table 3 gives the coefficients for numerical predictors of the final normalized model, and Table 4 gives the coefficients of the states in the final normalized model. All coefficients in tables 3 and 4 are significant at alpha = 0.05, Figure 1 shows the final model's residuals.

Table 1: Nested model comparisons

Smaller Model	Larger Model	F	p
Reduced	Full	0.905	0.501
Reduced log income	Full log income	0.905	0.404

Table 2: Fit statistics

Model	RMSE	MAE	R-squared
Full	1.4201	1.0273	0.6794
Reduced	1.4208	1.0278	0.6792
Full log income	1.4154	1.0218	0.6816
Reduced log income	1.4155	1.0219	0.6815

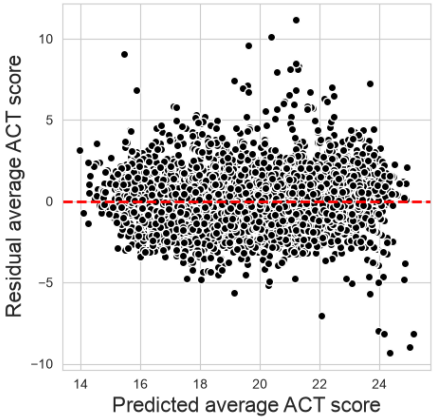
Table 3: Coefficients of the final model

Predictor	Coefficient	Standard Error
Intercept	16.6642	0.290
unemployment rate	-0.0883	0.023
percent college	0.4045	0.024
percent lunch	-1.8210	0.023
percent married	0.0897	0.025
log median income	-0.2295	0.031

Table 4: State coefficients of the final model (Reference: Delaware)

State	Coefficient	Standard Error	State	Coefficient	Standard Error
FL	3.9225	0.298	NJ	3.2696	0.300
GA	3.9140	0.299	NY	3.6155	0.303
IL	3.1542	0.296	OH	4.3492	0.295
IN	3.4886	0.300	PA	3.6214	0.296
KY	2.9137	0.307	TN	2.8711	0.303
LA	3.5278	0.308	TX	3.8488	0.294
MA	3.9424	0.303	WA	4.3350	0.303
MI	2.5090	0.297	WI	4.3066	0.300
MO	3.9071	0.300	WY	1.4724	0.370
NC	3.8231	0.299			

Figure 1: Residuals vs. final model's predicted average ACT score



Discussion

All five numerical predictors were significant in the final model, but their effects differed greatly. Free and reduced-price lunch dominated. Each standard deviation increase was associated with a 1.82 points lower average ACT score. This was about 4.5 times greater than the effect of percent college (0.40), about eight times that of log median income (-0.23) and about 20 times greater than that of unemployment and percent married. One possible explanation is that percent lunch describes the school's students while the others describe surrounding socioeconomic status. The negative coefficient on log median income is misleading. In the original model, median income was positively correlated with average ACT score and

not significant. Once logged, it and percent married became significant, suggesting their coefficients are dependent on each other.

Relative to Delaware, all state coefficients were significant, ranging from 1.47 in Wyoming to 4.35 in Ohio. This suggests location has an impact on school's average ACT score.

Neither school type nor charter status added explanatory value. There was a small amount of charter schools, only 170, and non-regular schools, only 12, so there is not enough data to make any conclusions on how the type of school affects average ACT scores.

The data are observational, so coefficients describe associations rather than causal effects. The models are evaluated on the same data used to fit them, so the reported statistics may overstate how well they would predict ACT scores for new schools.

Conclusions

This project asked how location, school type, and socioeconomic factors relate to the average ACT score of U.S. high schools. Using 7,227 schools from 20 states, a regression model with state, and five socioeconomic predictors explained about 68% of the variation in school average ACT scores.

Free and reduced-price lunch had by far the largest association with average ACT, followed by the percent of adults with a college degree. Unemployment rate, percent of children in married-couple families, and median income had much smaller associations. Each state had a significant association. School type and charter status added no detectable value, although the data contained very few schools that were not regular, non-charter schools.

The results support the idea that a school's average ACT score is closely tied to its location and socioeconomic factors. They describe associations in data from one year and not causal effects. Future work could improve on this project by testing the models on schools not used to fit them, include more schools that are not regular, non-charter schools, and include ACT participation rates.

References

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